Tarea Sección 10.1

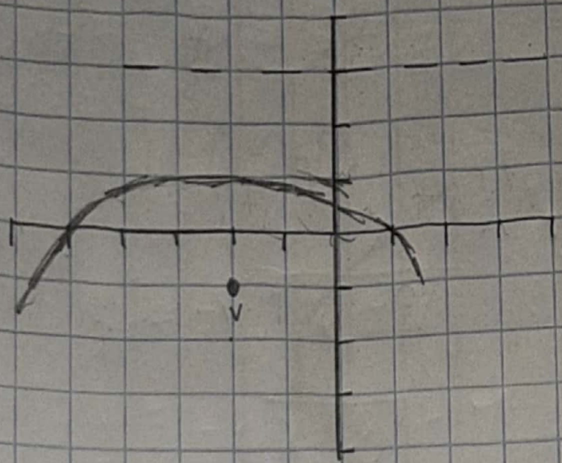
5) $(x+2)^2 = -8(y-1)$ $\quad 4P = -8$

$P = -2$

$V = (-2, 1)$

$F = (0+(-2), (1+(-2)) = (-2, -1)$

Directriz $= y = 2+(1) = y = 3$



11) $y^2 + 14y + 4x + 45 = 0$

$y^2 + 14y + 49 = -4x - 45 + 49$

$(y+7)^2 = -4x + 4$

$(y+7)^2 = -4(x-1)$

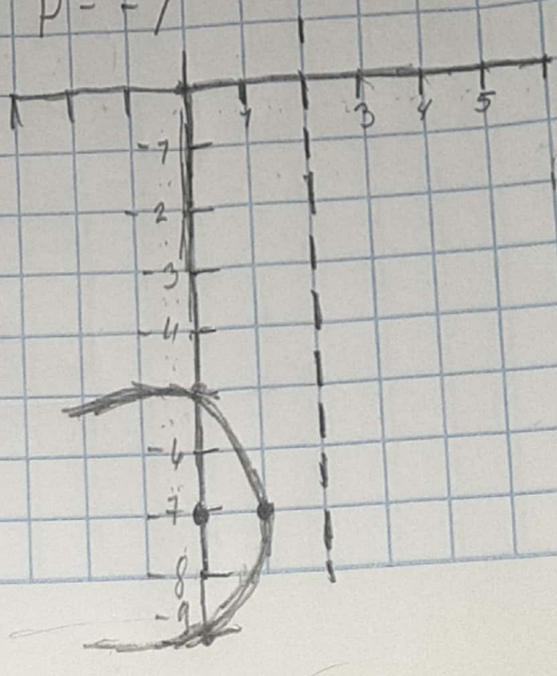
$V = (1, -7)$

$F = ((1-1), (7)) = F = (0, -7)$

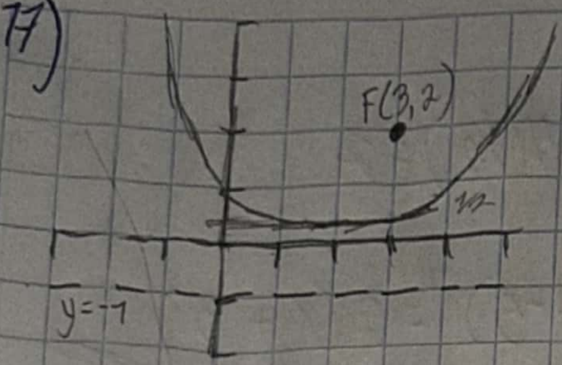
$D = x = 1-(-1) = x = 2$

$4P = -4 \rightarrow 1/2 =$ por cada lado

$P = -1$



17)



$V = (3, \frac{1}{2})$

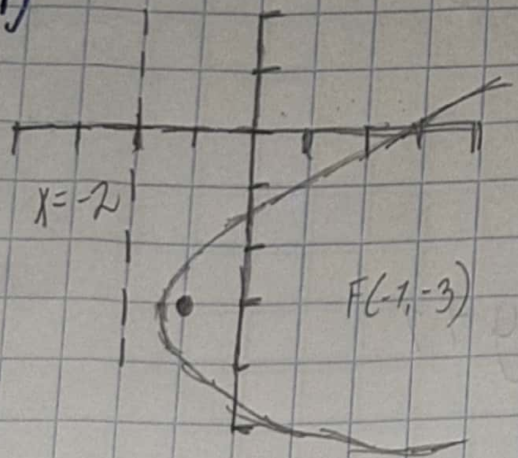
$(x-h)^2 = 4P(y-k)$

$(x-3)^2 = 4P(y - \frac{1}{2})$

$-1 = 4P + \frac{1}{2}$

$\underline{(x-3)^2 = 6(y - \frac{1}{2})}$

19)



$(y-k)^2 = 4P(x-h)$

$x = h - P$ $\qquad$ $4P = \frac{1}{2}$

$-2 = -\frac{3}{2} - P$ $\qquad$ $P = 2$

$V(-\frac{3}{2}, -3)$

$\underline{(y+3)^2 = 2(x + \frac{3}{2})}$

27) $V(-2, 3)$ $\qquad$ $y = 5$

$(x-h)^2 = 4P(y-k)$

$(x+2)^2 = 4P(y-3)$

$\underline{(x+2)^2 = -8(y-3)}$

$y = k - P$

$5 = 3 - P$

$P = 3 - 5$

$P = -2$

$4P$

$4(-2) = -8$

35) $V(-3,5)$ $\quad P(5,9)$

$(y-k)^2 = 4P(x-h)$ $\quad 4P = 4(1/2) = 2$

$(y-5)^2 = 4P(x+3)$

$(9-5)^2 = 4P(5-3)$ $\quad \underline{(y-5)^2 = 2(x+3)}$

$2 = 4P$ $\quad 4P = 2$ $\quad P = 1/2$

41) Mitad inferior $(y+1)^2 = x+3$

$y^2 = x+3-1$

$\underline{y = -\sqrt{x+3} - 1}$

47) Mitad izquierda $(x-2)^2 = y+1$

$x^2 = -(y+1)+2$

$\underline{x = -\sqrt{y+1} + 2}$

53) $P(2,5)$ $\quad Q(-2,-3)$ $\quad R(1,6)$

$\begin{cases} 4a+2b+c = 5 \\ -4a+2b-c = 3 \end{cases}$ $\quad 4b = 8$ $\quad b = 2$

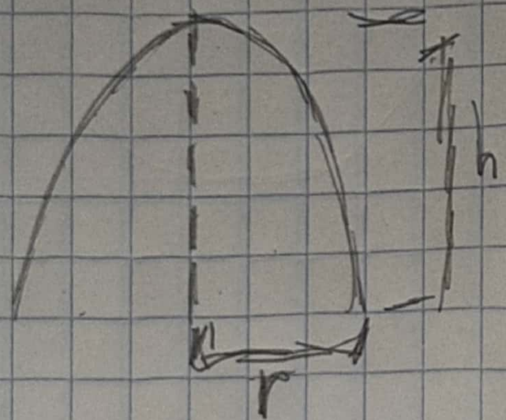
$5 = -4+4-C$ $\quad \underline{y = -x^2+2x+5}$

$C = 5$

$a = -1$ $\quad C = 5$

$b = 2$

63)



a) $x^2 = 4py$

$r^2 = 4ph$

$p = \frac{r^2}{4h}$

b) $V(0,0)$ $\quad y - 10 = 0$

$F(0,10)$

$\sqrt{(x_1 - x_2)^2 + (y_1 + y_2)^2}$

$\sqrt{(x_1)^2 + (5-10)^2} = 15$

$x_1 = \sqrt{225 - 25}$

$x = \sqrt{200} = 10\sqrt{2}$ pies